

SQL QUERIES

1. Displaying details of HOD of all the Departments.

Query:

```
select dept_name AS "DEPARTMENT" , fac_name AS "HOD" ,  
TIMESTAMPDIFF(YEAR,DOB,CURDATE()) AS "AGE" , address AS "CITY" , experience AS  
"EXPERIENCE" from hod h, faculty f, dept d where h.hod_id=f.fac_id and d.dept_id=h.dept_id ;
```

Output:

DEPARTMENT	HOD	AGE	CITY	EXPERIENCE
Computer Science and Engineering	MANINDER SINGH	48	PATIALA	17
Electronics and Communication Engineering	ALPANA AGGARWAL	51	PATIALA	17
Electrical and Instrumentation Engineering	RS KALER	51	LUDHIANA	17
Mechanical Engineering	TARUN KUMAR BERA	38	KARNAL	8
Chemical Engineering	RAJEEV MEHTA	55	LUDHIANA	22

2. Displaying No of Students in a Particular Hostel.

Query:

```
select h.hostel_id AS "HOSTEL ID", CONCAT("HOSTEL - ",h.hostel_name) AS "HOSTEL  
NAME" ,h.category AS "GIRLS/BOYS" , count(*) AS "TOTAL RESIDENTS" from student s ,  
hostel h , allocated_hostel a where h.hostel_id=a.hostel_id and s.roll_no=a.roll_no group by  
h.hostel_id,h.hostel_name,h.category order by h.hostel_id;
```

Output:

HOSTEL ID	HOSTEL NAME	GIRLS/BOYS	TOTAL RESIDENTS
1	HOSTEL - M	BOYS	9
2	HOSTEL - O	BOYS	9
3	HOSTEL - K	BOYS	8
4	HOSTEL - N	GIRLS	8
5	HOSTEL - G	GIRLS	11

3. Displaying Faculty ID, Faculty Name, Designation , Salary of Faculties whose Monthly salary is greater than or equal to 200000.

Query:

```
select f.fac_id AS "FACULTY ID", f.fac_name "FACULTY NAME",UPPER(g.grade_name) as  
"DESIGNATION", (1+f.experience/100)*(g.basic_pay+g.hra+g.da) AS "MONTHLY SALARY" from  
faculty f , grade_pay g where f.grade_id=g.grade_id and  
(1+f.experience/100)*(g.basic_pay+g.hra+g.da)>=200000 order by fac_id;
```

Output:

FACULTY ID	FACULTY NAME	DESIGNATION	MONTHLY SALARY
1	RK SHARMA	PROFESSOR	260325.0000
2	SEEMA BAWA	PROFESSOR	253650.0000
3	RAJESH KUMAR	ASSOCIATE PROFESSOR	209000.0000
4	MANINDER SINGH	PROFESSOR	260325.0000
5	INDERVEER CHANNA	ASSOCIATE PROFESSOR	222300.0000
6	SHALINI BATRA	ASSOCIATE PROFESSOR	207100.0000
11	ALPANA AGGARWAL	PROFESSOR	260325.0000
12	AMIT KUMAR KOHLI	ASSOCIATE PROFESSOR	216600.0000
17	RS KALER	PROFESSOR	260325.0000
18	MUKESH SINGH	ASSOCIATE PROFESSOR	207100.0000
23	AJAY BATISH	PROFESSOR	267000.0000
24	SK MOHAPATRA	ASSOCIATE PROFESSOR	224200.0000
29	RAJEEV MEHTA	PROFESSOR	271450.0000
30	HARIPADA BHUNIA	ASSOCIATE PROFESSOR	226100.0000
35	PRAKASH GOPLAN	PROFESSOR	280350.0000
36	GURBINDER SINGH	REGISTRAR	291600.0000
37	RUCHI GUPTA	REGISTRAR	279450.0000
38	AMANPREET SINGH	GAZZETTED OFFICER	208050.0000
39	PANKAJ SINHA	GAZZETTED OFFICER	211700.0000

4. Displaying Total Staff in a Particular Department.

Query:

```
select d.dept_id AS "DEPARTMENT ID", d.dept_name AS "DEPARTMENT NAME" , count(*) AS
"TOTAL STAFF" from teaching_fac f, dept d where f.dept_id=d.dept_id group by
d.dept_id,d.dept_name;
```

Output:

DEPARTMENT ID	DEPARTMENT NAME	TOTAL STAFF
1	Computer Science and Engineering	10
2	Electronics and Communication Engineering	6
3	Electrical and Instrumentation Engineering	6
4	Mechanical Engineering	6
5	Chemical Engineering	6

5. Displaying Name of Faculty of a Particular Department who is having maximum work Experience in that Department.

Query:

```
select d.dept_id AS "DEPARTMENT ID", d.dept_name AS "DEPARTMENT NAME" ,f.fac_name
AS "NAME",f.experience AS "EXPERIENCE" from faculty f,teaching_fac t, dept d where
t.dept_id=d.dept_id and f.fac_id=t.fac_id and (d.dept_id,f.experience) in (select d.dept_id ,
max(f.experience) from dept d , faculty f , teaching_fac t where d.dept_id=t.dept_id and
t.fac_id=f.fac_id group by d.dept_id);
```

Output:

DEPARTMENT ID	DEPARTMENT NAME	NAME	EXPERIENCE
1	Computer Science and Engineering	RK SHARMA	17
1	Computer Science and Engineering	MANINDER SINGH	17
1	Computer Science and Engineering	INDERVEER CHANNA	17
2	Electronics and Communication Engineering	ALPANA AGGARWAL	17
3	Electrical and Instrumentation Engineering	RS KALER	17
4	Mechanical Engineering	AJAY BATISH	20
5	Chemical Engineering	RAJEEV MEHTA	22

6. Displaying details of Students who are day scholars along with the course in which they are enrolled.

Query:

```
select s.roll_no AS "ROLL NO" , s.name AS "NAME" ,
TIMESTAMPDIFF(YEAR,s.DOB,CURDATE()) AS "AGE", s.address AS "CITY" , s.phone_no AS
"PHONE NO" , c.course_name AS "COURSE" from student s, course c where s.roll_no not in
(select a.roll_no from allocated_hostel a) and c.course_id=s.course_id order by s.roll_no;
```

Output:

ROLL NO	NAME	AGE	CITY	PHONE NO	COURSE
101801871	Akshat Khosla	22	Kolkata	7108025987	BE MECH ENGG
102003001	Kushagar Bansal	20	Patiala	9336271289	BE COE
102003004	Deeksha Aggarwal	19	Delhi	7136279289	BE COE
102102947	Akshita Garg	18	Faridabad	9412015310	BE MECH ENGG
102103102	Sanidhya Vijay	19	Dehradun	9876511289	BE COE
102103103	Shashank Singh	21	Jammu	9753112345	BE COE
111905101	Harsh Gupta	20	Rudrapur	7178479212	BE COE
121805303	Ayush Modi	22	Moradabad	7212345113	BE COE
131805000	Anuj Gupta	18	Lucknow	9318275149	BE CSE
131903100	Aryan Kapoor	22	Patna	7674290289	BE CSE
132005100	Dhruv Mishra	20	Ambala	7052191499	BE CSE
132105301	Diya Gupta	19	Ranchi	9651000212	BE CSE
141805001	Kartik Sood	23	Ranchi	8136217626	BE COBS
141901232	Devansh Garg	21	Karnatka	90012912282	BE CHEM ENGG
141903100	Shiti Garg	21	Patna	9514299489	BE COBS
142002178	Khushi Garg	20	Bangalore	9678291009	BE CHEM ENGG
142103221	Rohit Bisht	19	Jodhpur	9619151200	BE CHEM ENGG
142105300	Riya Kohli	18	Ambala	9712111300	BE COBS
152004718	Kunal Madaan	24	Agra	9108113123	ME CHEM ENGG
171903100	Shrey	22	Agra	9004290181	BE ECE
172005100	Chahat Goyal	21	Kanpur	7098290169	BE ECE
172105300	Ananya Singh	18	Faridabad	9712015311	BE ECE
182103100	Neha Aggarwal	24	Agra	9014290132	ME ECE
182109117	Naina Madaan	23	Agra	9014970230	ME POWER SY...
192105300	Uday Uppal	18	Faridabad	9612015300	BE EIC

7. Displaying details of first 10 Students and their course who have highest CGPA.

Query:

```
select s.roll_no AS "ROLL NO" , s.name AS "NAME" , c.course_name AS "COURSE" , r.cgpa
from student s, course c ,result r where s.roll_no=r.roll_no and c.course_id=s.course_id and
r.semester=(s.semester-1) order by r.cgpa desc LIMIT 10;
```

Output:

ROLL NO	NAME	COURSE	cgpa
102103100	Ashish Kakkar	BE COE	10
111905100	Jatin Garg	BE COE	10
161805000	Anshita	BE ENC	9.99
101904970	Taasha	BE MECH ENGG	9.98
152004718	Kunal Madaan	ME CHEM ENGG	9.98
102103101	Karan Nasa	BE COE	9.85
142105300	Riya Kohli	BE COBS	9.63
102003004	Deeksha Aggarwal	BE COE	9.61
131903100	Aryan Kapoor	BE CSE	9.54
132105301	Diya Gupta	BE CSE	9.43

8. Displaying No of students from a particular city.

Query:

```
select address AS "CITY" ,count(*) AS "TOTAL STUDENTS" from student group by address
order by count(*) desc;
```

Output:

CITY	TOTAL STUDENTS
Ambala	8
Agra	6
Delhi	6
Patna	5
Ranchi	5
Kolkata	4
Ludhiana	4
Kanpur	3
Faridabad	3
Rudrapur	3
Patiala	2
Jalandhar	2
Lucknow	2
Karnatka	2
Bangalore	2
Jodhpur	2
Pune	2
Amritsar	1
Karnal	1
Chandigarh	1
Dehradun	1
Jammu	1
Gurgaon	1
Moradabad	1
Mumbai	1

9. Displaying all the Graduate courses offered by the university and their Corresponding Department.

Query:

```
select c.course_name AS "COURSE NAME" , d.dept_name AS "DEPARTMENT NAME" from
dept d, course c where d.dept_id=c.dept_id and c.course_name like "BE%";
```

Output:

COURSE NAME	DEPARTMENT NAME
BE COE	Computer Science and Engineering
BE CSE	Computer Science and Engineering
BE COBS	Computer Science and Engineering
BE ENC	Electronics and Communication Engineering
BE ECE	Electronics and Communication Engineering
BE EIC	Electrical and Instrumentation Engineering
BE MECH ENGG	Mechanical Engineering
BE MECHATRONICS	Mechanical Engineering
BE CHEM ENGG	Chemical Engineering

10. Displaying all the Post Graduate courses offered by the university and their Corresponding Department.

Query:

```
select c.course_name AS "COURSE NAME" , d.dept_name AS "DEPARTMENT NAME" from
dept d, course c where d.dept_id=c.dept_id and c.duration = 2;
```

Output:

COURSE NAME	DEPARTMENT NAME
ME SOFT ENGG	Computer Science and Engineering
ME ECE	Electronics and Communication Engineering
ME POWER SYSTEMS	Electrical and Instrumentation Engineering
ME CAD/CAM	Mechanical Engineering
ME CHEM ENGG	Chemical Engineering

11. Displaying details of DEANS of the university.

Query:

```
select d.designation AS "DESIGNATION", f.fac_name AS "NAME", d.email_id AS "EMAIL ID",
d.office_contact AS "CONTACT NO." from deans d,faculty f where d.dean_id=f.fac_id;
```

Output:

DESIGNATION	NAME	EMAIL ID	CONTACT NO.
Dean of Research and Sponsered Projects	RAJESH KUMAR	dorsp@thapar.edu	2393038
Dean of Academic Affairs	MANINDER SINGH	doaa@thapar.edu	2393022
Dean of Student Affairs	INDERVEER CHANNA	dosa@thapar.edu	2393039
Dean of Faculty Affairs	RS KALER	dofa@thapar.edu	2393914

12. Displaying details of Directorates of the university.

Query:

```
select upper(d.designation) AS "DESIGNATION", f.fac_name AS "NAME", d.email_id AS
"EMAIL ID", d.office_contact AS "CONTACT NO." from directorate d,faculty f where
d.dir_id=f.fac_id;
```

Output:

DESIGNATION	NAME	EMAIL ID	CONTACT NO.
DEPUTY DIRECTOR	AJAY BATISH	deputydirector@thapar.edu	2393521
DIRECTOR	PRAKASH GOPLAN	director@thapar.edu	2393001

13. Displaying details of Non-Teaching Staff of the University.

Query:

```
select f.fac_id AS "FACULTY ID", f.fac_name AS "NAME", t.designation AS "DESIGNATION",
TIMESTAMPDIFF(YEAR,f.DOB,CURDATE()) AS "AGE" , f.address AS "CITY" , f.experience AS
"EXPERIENCE" from faculty f, non_teaching_fac t where t.fac_id=f.fac_id;
```

Output:

FACULTY ID	NAME	DESIGNATION	AGE	CITY	EXPERIENCE
36	GURBINDER SINGH	REGISTRAR	52	KAPURTHALA	20
37	RUCHI GUPTA	ASSISTANT REGISTRAR	47	LUDHIANA	15
38	AMANPREET SINGH	LAW OFFICER	43	SIRHIND	14
39	PANKAJ SINHA	FINANCE OFFICER	49	PATHANKOT	16
40	ASHISH MEHRA	SENIOR MANAGER (AUDITS)	42	HYDERABAD	12
41	P.S. JOLLY	SENIOR MANAGER (S&P)	45	PANIPAT	15
42	VINEET ARORA	ACCOUNTS MANAGER	40	MANDI	11

14. Displaying Subjects of the course BE-ENC.

Query:

```
select s.sub_id AS "SUBJECT ID", s.sub_name "SUBJECT NAME" from course c , subject s ,
course_subject cs where c.course_id=cs.course_id and s.sub_id=cs.sub_id and c.course_id =
(select course_id from course where course_name = 'BE ENC');
```

Output:

SUBJECT ID	SUBJECT NAME
1	Applied Chemistry
2	Computer Programming
3	Electrical Engineering
4	Energy and Environment
5	Mathematics-I
6	Mechanics
7	Applied Physics
8	Object Oriented Programming
9	Electronics Engineering
10	Engineering Drawing
11	Professional Communication
12	Mathematics-II
13	Data Structures
19	Project Semester
20	Machine Learning
25	Optimisation Process
26	Operating Systems
27	Capstone Project

15. Displaying Roll no and Name of Students residing in AC and 2-S room of Hostel with their Hostel Names.

Query:

```
select s.roll_no AS "ROLL NO" , upper(s.name) AS "NAME" , concat('HOSTEL - ',h.hostel_name)
AS "HOSTEL" from student s , allocated_hostel a , hostel h where a.roll_no=s.roll_no and
a.hostel_id=h.hostel_id and a.room_type='AC' and a.room_occupancy='2S';
```

Output:

ROLL NO	NAME	HOSTEL
102003002	KIRTI KAPOOR	HOSTEL - N
102003005	VIDUL GUPTA	HOSTEL - M
121805300	PRACHI GUPTA	HOSTEL - G
121805301	KAMYA TAYAL	HOSTEL - N
132005002	INAYAT GOYAL	HOSTEL - N
142005100	ARPIT SAGAR	HOSTEL - O
152005001	SIMRAN KAUR	HOSTEL - G
152102132	DIVYA GARG	HOSTEL - N
162105300	GIRIK GARG	HOSTEL - M
191805000	TARIK BHATEJA	HOSTEL - O

16. Displaying Department and where they are located.

Query:

```
select dept_name AS "DEPARTMENT" , location AS "LOCATION" from dept;
```

Output:

DEPARTMENT	LOCATION
Computer Science and Engineering	LC
Electronics and Communication Engineering	B Block
Electrical and Instrumentation Engineering	C Block
Mechanical Engineering	F Block
Chemical Engineering	E Block

17. Displaying credits of the courses along with their departments.

Query:

```
select c.course_id AS "COURSE ID" , c.course_name AS "COURSE NAME" , d.dept_name AS
"DEPARTMENT", c.credits AS "CREDITS" from course c , dept d where c.dept_id=d.dept_id;
```

Output:

COURSE ID	COURSE NAME	DEPARTMENT	CREDITS
1	BE COE	Computer Science and Engineering	220
2	BE CSE	Computer Science and Engineering	225
3	BE COBS	Computer Science and Engineering	195
4	ME SOFT ENGG	Computer Science and Engineering	120
5	BE ENC	Electronics and Communication Engineering	220
6	BE ECE	Electronics and Communication Engineering	220
7	ME ECE	Electronics and Communication Engineering	110
8	BE EIC	Electrical and Instrumentation Engineering	195
9	ME POWER SYSTEMS	Electrical and Instrumentation Engineering	115
10	BE MECH ENGG	Mechanical Engineering	210
11	BE MECHATRONICS	Mechanical Engineering	205
12	ME CAD/CAM	Mechanical Engineering	120
13	BE CHEM ENGG	Chemical Engineering	185
14	ME CHEM ENGG	Chemical Engineering	110

18. Displaying Wardens of the Hostels with their capacity and category .

Query:

```
select concat('HOSTEL - ',h.hostel_name) AS "HOSTEL NAME" , f.fac_name "WARDEN" ,
h.capacity AS "CAPACITY", h.category AS "CATEGORY" from hostel h , faculty f where
h.warden_id=f.fac_id;
```

Output:

HOSTEL NAME	WARDEN	CAPACITY	CATEGORY
HOSTEL - M	RAJEEV MEHTA	1000	BOYS
HOSTEL - O	RAJESH KUMAR	900	BOYS
HOSTEL - K	AMIT MISHRA	300	BOYS
HOSTEL - N	SWATI SONDEHI	800	GIRLS
HOSTEL - G	NEETU SINGH	200	GIRLS

19. Displaying count of Staff members who have particular degree.

Query:

```
select qualification AS "QUALIFICATION",count(*) AS "NO. OF TEACHERS" from teaching_fac
group by qualification;
```

Output:

QUALIFICATION	NO. OF TEACHERS
PHD	15
DOCTORATE	9
M TECH	6
B TECH	4

20. Displaying count of students enrolled in specific course.

Query:

```
select c.course_name AS "COURSE NAME",count(*) As 'NO OF STUDENTS' from student s ,
course c where s.course_id = c.course_id group by c.course_name;
```

Output:

COURSE NAME	NO OF STUDENTS
BE COE	18
BE CSE	8
BE COBS	8
ME SOFT ENGG	4
BE ENC	4
BE ECE	4
ME ECE	2
BE EIC	4
ME POWER SYSTEMS	2
BE MECH ENGG	4
BE MECHATRONICS	4
ME CAD/CAM	2
BE CHEM ENGG	4
ME CHEM ENGG	2

21. Displaying top 5 Faculties who has minimum Experience.

Query:

```
select f.fac_id AS "FACULTY ID", f.fac_name "FACULTY NAME",UPPER(g.grade_name) as
"DESIGNATION" ,f.experience AS "EXPERIENCE" from faculty f , grade_pay g ,teaching_fac t
where f.grade_id=g.grade_id and f.fac_id=t.fac_id order by f.experience LIMIT 5;
```

Output:

FACULTY ID	FACULTY NAME	DESIGNATION	EXPERIENCE
28	NAVDEEP KUMMAR	LECTURER	4
27	SANDEEP KUMAR SHA...	ASSISTANT PROFESSOR-I	5
9	ANSHU PRASHAR	ASSISTANT PROFESSOR-I	5
10	SAHIL SHARMA	LECTURER	5
22	NAVDEEP KAUR	LECTURER	5

22. Displaying Faculty Members who are both HOD and DEAN of the University.

Query:

```
select f.fac_id AS "FACULTY ID", f.fac_name "FACULTY NAME",UPPER(g.grade_name) as
"DESIGNATION" ,f.experience AS "EXPERIENCE" from faculty f , grade_pay g ,hod h,deans d
where f.grade_id=g.grade_id and f.fac_id=h.hod_id and d.dean_id=f.fac_id;
```

Output:

FACULTY ID	FACULTY NAME	DESIGNATION	EXPERIENCE
4	MANINDER SINGH	PROFESSOR	17
17	RS KALER	PROFESSOR	17

23. Displaying Average CGPA of the Semesters.

Query:

```
select semester AS "SEMESTER", ROUND(avg(sgpa),2) AS "AVERAGE SGPA" from result
group by semester;
```

Output:

SEMESTER	AVERAGE SGPA
7	7.78
5	8.04
3	7.91
1	8.66

24. Displaying count of Male and Female students studying in the University.

Query:

```
select CASE WHEN gender='M' THEN 'MALE' WHEN gender='F' THEN 'FEMALE' END AS
"GENDER" , count(*) AS "NO. OF STUDENTS" from student group by gender;
```

Output:

GENDER	NO. OF STUDENTS
MALE	41
FEMALE	29

25. Displaying count of Male and Female staff working in the University.

Query:

```
select CASE WHEN gender='M' THEN 'MALE' WHEN gender='F' THEN 'FEMALE' END AS
"GENDER" , count(*) AS "COUNT OF FACULTY MEMBERS" from faculty group by gender;
```

Output:

GENDER	COUNT OF FACULTY MEMBERS
MALE	30
FEMALE	12